

MUHAMMAD TAAHA

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EDUCATION

National University of Computer and Emerging Sciences (FAST-NUCES) Karachi, Pakistan
Bachelor of Science in Computer Science; GPA: 3.47/4.00 2023 – 2027

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, SQL, C/C++, Java, x86 Assembly
- **Frameworks/Tools:** React.js, Node.js, Express.js, Flask, PostgreSQL, Docker, AWS, Git, Linux, Claude Code
- **Concepts:** Distributed Systems (CRDTs), Network Programming, REST APIs, n8n Workflow Automation, AI Workflow Integration
- **Certifications:** Python Fundamentals (Microsoft), Data Analysis (IBM)

EXPERIENCE

- **Coders Cup 2025** Nov 2025
 - Led a team of 30+ organizers to execute a nationwide competitive programming event with 1,000+ participants.
 - Managed contest infrastructure and ensured smooth execution of real-time submissions and leaderboard updates.
- **DWT: Data Structures With Taaha** Jun 2025 – Jul 2025
 - Founded and led a 4-week data structures program covering arrays, linked lists, stacks, trees, and hashing.
 - Conducted 12+ live sessions for 100+ students, focusing on problem-solving and interactive learning.
 - Designed curriculum, curated coding problems, and organized quizzes and contests to reinforce concepts.

PROJECTS

- **P3883: Unary Postfix Operator for Boolean Inversion in C++** | ISO C++ WG21 Oct 2025
 - Authored a formal ISO C++ language proposal submitted to WG21 (Library Evolution Group) introducing a postfix `flag~` operator for in-place boolean toggling.
 - Designed the operator to be fully backward-compatible, with void return type and restriction to bool lvalues, preventing misuse and chaining.
 - Evaluated and documented alternatives (XOR assignment, `std::flip`, prefix reuse, new keywords) with rationale for the postfix approach.
- **Stack: Digital Social Card App** | Play Store Feb 2026
 - Built and deployed a production-ready Android application for selective QR-based sharing of social profiles in live networking settings.
 - Developed a mobile-first cross-platform stack using React, TypeScript, Capacitor, and Vite with an emphasis on speed and responsiveness.
 - Implemented dynamic QR generation, platform-aware deep-link validation, and offline-first local persistence without account dependency.
- **p2pdocs: Decentralized Real-time Collaborative Editor** | GitHub Nov 2025
 - Engineered a serverless collaborative editing system using **CRDT (RGA)** algorithms to ensure state convergence across decentralized nodes.
 - Built a thread-safe hybrid networking layer with **UDP/TCP** for peer discovery and low-latency synchronization on local networks.
 - Published a zero-dependency package on **PyPI** with locking mechanisms for concurrent UI and network operations.
- **Guardian: Mission-Critical Student Safety Portal** | GitHub Aug 2025
 - Developed a secure communication hub for 400+ users, integrating Role-Based Access Control (RBAC) for structured access management.
 - Utilized the **Google Sheets API** as a live backend for real-time updates without redeployment.
 - Engineered an admin-controlled "Emergency Mode" to override privacy filters and expose critical details during crises.